EECS 127/227A – Final Review Summary

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1 Linear Algebra – Vectors

1.1 Vector Space, Subspace, Affine Set, and Basis

- $\mathbb{R}^n$ is the space of vectors with n elements.
- Vectors $x^{(1)}, \ldots, x^m \in \mathbb{R}^n$ are linearly dependent if there is a non-trivial linear combination $\sum_i \alpha_i x^{(i)}$ which is the zero vector. Otherwise, they are linearly independent.
- A non-empty set $\mathcal{S} \subseteq \mathbb{R}^n$ is a subspace if for all $x, y \in \mathcal{S}$ and scalars α, β we have $\alpha x + \beta y \in \mathcal{S}$.
- For m vectors $x^{(1)}, \ldots, x^m \in \mathbb{R}^n$, we define span$(x^{(1)}, \ldots, x^m)$ as the set of all linear combinations of $x^{(1)}, \ldots, x^m$. This set is a subspace.
- A set of vectors $x^{(1)}, \ldots, x^{(d)}$ is a basis for a subspace $\mathcal{S}$ if
  - $x^{(1)}, \ldots, x^{(d)}$ are linearly independent;
  - For all $x \in \mathcal{S}$, there exist scalars $\alpha_1, \ldots, \alpha_d$ such that $x = \sum_i \alpha_i x^{(i)}$.
- For a subspace $\mathcal{S}$, the basis is not unique, but all bases have the same number of vectors, d. This number d is the dimension of the subspace $\mathcal{S}$.
- A set $\mathcal{X} \subseteq \mathbb{R}^n$ is affine if there is a subspace $\mathcal{S} \subseteq \mathbb{R}^n$ and a vector $x^{(0)} \in \mathbb{R}^n$ such that $\mathcal{X} = x^{(0)} + \mathcal{S}$ (adding $x^{(0)}$ to all vectors in $\mathcal{S}$). To prove a set to be affine, first find $x^{(0)}$ and then show that $\mathcal{X} - x^{(0)}$ is a subspace.

1.2 Inner Product and Orthogonal Vectors

- For a pair of vectors $x, y \in \mathbb{R}^n$, the standard inner product (dot product) is $\langle x, y \rangle = x^\top y = y^\top x = x_1 y_1 + \cdots + x_n y_n$.
- It holds that $\langle x, y \rangle = \|x\|_2 \|y\|_2 \cos(\theta)$, where θ is the angle between x and y.
- Two vectors are orthogonal if $\langle x, y \rangle = 0$ (denoted $x \perp y$).
- d vectors $x^{(1)}, \ldots, x^{(d)}$ are mutually orthogonal if $x^{(i)} \perp x^{(j)}$ for all $i \neq j$. This guarantees that $x^{(1)}, \ldots, x^{(d)}$ are linearly independent.
- We say $x^{(1)}, \ldots, x^{(d)}$ are orthonormal if they are mutually orthogonal and have norm one. I.e., $\|x^{(i)}\|_2^2 = \langle x^{(i)}, x^{(i)} \rangle = 1$ for all $i = 1, \ldots, d$ and $\langle x^{(i)}, x^{(j)} \rangle = 0$ for all $i \neq j$.

1.3 Vector Norms

- A function $\|\cdot\| : \mathbb{R}^n \to \mathbb{R}$ is a norm if
  1. $\|x\| \geq 0$ for all $x \in \mathbb{R}^n$ and $\|x\| = 0$ if and only if $x = 0$;
  2. $\|x + y\| \leq \|x\| + \|y\|$ for all $x, y \in \mathbb{R}^n$;
  3. $\|\alpha x\| = |\alpha| \|x\|$ for all $\alpha \in \mathbb{R}, \forall x \in \mathbb{R}^n$.
- An ℓ_p norm, for $1 \leq p < \infty$, is of the form $\|x\|_p = (|x_1|^p + \cdots + |x_n|^p)^{1/p}$.
- We define $\|x\|_0$ to be the number of non-zero elements in x. This is not a true norm but appears frequently.
- For an arbitrary vector $x \in \mathbb{R}^n$, it holds that $\|x\|_2^2 = x^\top x$.

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1.4 Linear Functions and Affine Functions

- A function $f(x) : \mathbb{R}^n \to \mathbb{R}$ is linear if $f(ax + by) = af(x) + bf(y)$ for all $x, y \in \mathbb{R}^n$ and scalars a, b.
- If $f(x)$ is linear, there exists an $a \in \mathbb{R}^n$ such that $f(x) = a^\top x$.
- A function $f(x)$ is affine if $f(x) - f(0)$ is linear. This means $f(x) = a^\top x + b$ for some $a \in \mathbb{R}^n$ and $b \in \mathbb{R}$.

1.5 Hyperplanes

- A hyperplane in $\mathbb{R}^n$ is a $(n-1)$ dimensional affine set, can be written as $H = \{z \in \mathbb{R}^n \mid a^\top z = b\}$ for a non-zero vector $a \in \mathbb{R}^n$ and scalar b.
- a is called the normal vector of the hyperplane. I.e., for any two vectors $z^1, z^2 \in H$, we have that $a \perp (z^1 - z^2)$.
- Hyperplanes divide $\mathbb{R}^n$ into half spaces $H_- = \{x \mid a^\top x \leq b\}$ and $H_+ = \{x \mid a^\top x \geq b\}$.

1.6 Projections

- Let $\mathcal{S}$ be a subspace of a space $\mathcal{X}$. The projection of a point $x \in \mathcal{X}$ onto $\mathcal{S}$ is $\Pi_{\mathcal{S}}(x) = \arg\min_{y \in \mathcal{S}} \|y - x\|$.
- The minimizer $y^\star = \Pi_{\mathcal{S}}(x)$ exists and is unique. Furthermore, $y^\star = \Pi_{\mathcal{S}}(x)$ if and only if $(x - y^\star) \perp \mathcal{S}$.
  I.e., $(x - y^\star)$ is orthogonal to every vector in $\mathcal{S}$.
- For projection onto an affine space, this condition becomes $(x - y^\star) \perp (y - y^\star)$ for all $y \in \mathcal{S}$.
  - Suppose that $y^{(1)}, \ldots, y^{(d)}$ form a basis for the affine space $\mathcal{S}$.
    We can find $y^\star$ by solving for the set of equations $y^\star \in \mathcal{S}$ and $y - y^\star \perp y^{(i)}$ for $i = 1, \ldots, d$.
- For projection onto a 1-dimensional subspace $\mathcal{S} = \text{span}(v)$, we have the formula $\Pi_{\mathcal{S}}(x) = \frac{\langle x, v \rangle}{\|v\|^2} v$.
- Now generalize projection onto a subspace $\mathcal{S} = \text{span}(x^{(1)}, \ldots, x^{(d)})$, where $x^{(1)}, \ldots, x^{(d)}$ are an orthonormal basis: $\Pi_{\mathcal{S}}(x) = \sum_i \langle x, x^{(i)} \rangle x^{(i)}$.

2 Linear Algebra – Matrices

2.1 Range, Nullspace, and Rank

- The range of A is the set of all linear combinations of A's columns: $\mathcal{R}(A) = \{Ax \mid x \in \mathbb{R}^n\}$.
- $\mathcal{R}(A)$ is a subspace, and its dimension is $\text{Rank}(A)$, which is equal to the number of linearly independent columns of A, and equal to the number of linearly independent rows.
- The nullspace of A is $\mathcal{N}(A) = \{x \in \mathbb{R}^n \mid Ax = 0\}$.
- The nullspace is also a subspace. The fundamental theorem of linear algebra relates the null space and the range:
  1. $\mathcal{N}(A) \perp \mathcal{R}(A^\top)$;
  2. $\mathcal{N}(A) \oplus \mathcal{R}(A^\top) = \mathbb{R}^n$, where $\oplus$ denotes "direct sum". I.e., any vector in $\mathbb{R}^n$ can be decomposed into a sum of a vector from the null space of A and a vector from the column space of $A^\top$;
  3. $\dim(\mathcal{N}(A)) + \text{Rank}(A) = n$.

2.2 Eigenvalues and Eigenvectors

- Consider a square matrix $A \in \mathbb{R}^{n \times n}$. If there exists a scalar λ and a vector v such that $Av = \lambda v$, then we say that λ is an eigenvalue of A and v is the corresponding eigenvector.
- To find the eigenvalues of A, we solve for λ that makes $\det(A - \lambda I) = 0$.
  Then, for each eigenvalue λ_i, we can solve $Av^{(i)} = \lambda_i v^{(i)}$ to find the corresponding eigenvector $v^{(i)}$.

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- If A is rank-deficient (not full rank, i.e., there are linear dependent rows/columns), then its determinant is 0 and at least one of its eigenvalues is 0.
- $AA^\top$ and $A^\top A$ share the same non-zero eigenvalues.
- A's trace (sum of the diagonal entries) is equal to the sum of its eigenvalues.

2.3 Symmetric Matrices and Positive/Negative (Semi)Definite Matrices

- A square matrix $A \in \mathbb{R}^{n\times n}$ is symmetric if $A = A^\top$. We denote the set of all $n \times n$ symmetric matrices as $\mathbb{S}^n$.
- The eigenvalues of a symmetric matrix are all real.
- A symmetric matrix $A \in \mathbb{S}^n$ is positive semidefinite (PSD, denoted as $A \succeq 0$ or $A \succcurlyeq 0$) if all eigenvalues are non-negative, i.e., $\lambda_1(A), \ldots, \lambda_n(A) \geq 0$. We denote the set of $n \times n$ PSD matrices as $\mathbb{S}^n_+$.
- An alternative PSD definition: A matrix $A \in \mathbb{S}^n$ is PSD if the scalar $x^\top Ax$ is non-negative for all $x \in \mathbb{R}^n$.
- Note: showing that all elements of a matrix are non-negative does NOT prove PSD.
- A symmetric matrix $A \in \mathbb{S}^n$ is positive definite (PD, denoted as $A \succ 0$) if all eigenvalues are strictly positive, i.e., $\lambda_1(A), \ldots, \lambda_n(A) > 0$. We denote the set of $n \times n$ PD matrices as $\mathbb{S}^n_{++}$.
  Alternatively, A is PD if $x^\top Ax > 0$ for all $x \neq 0$.
- An easier way to check whether a matrix is PD without calculating eigenvalues: A symmetric matrix A is PD if and only if all of its leading principal minors are strictly positive.
- A symmetric matrix A is negative semidefinite (NSD) if $\lambda_1(A), \ldots, \lambda_n(A) \leq 0$ or $x^\top Ax \leq 0$ for all $x \in \mathbb{R}^n$.
- A symmetric matrix A is negative definite (ND) if $\lambda_1(A), \ldots, \lambda_n(A) < 0$ or $x^\top Ax < 0$ for all $x \neq 0$.
- All PD matrices are PSD and all ND matrices are NSD.
- A matrix neither PSD nor NSD is called sign indefinite. It has at least one positive and one negative eigenvalue.

2.4 Orthogonal Matrices

- A square matrix $U \in \mathbb{R}^{n\times n}$ with columns $u^{(1)}, \ldots, u^{(n)}$ is called orthogonal if its columns are orthonormal to each other. I.e., the columns are mutually orthogonal and have norm 1.
  I.e., for arbitrary pairs of $i, j \in \{1, \ldots, n\}$, we have $\langle u^{(i)}, u^{(j)}\rangle$ is 1 if $i = j$ and 0 if $i \neq j$.
- A matrix U is orthogonal if and only if $U^\top U = I_n$, where I_n denotes the $n\times n$ identity matrix. I.e., $U^\top = U^{-1}$.
- An identity matrix is orthogonal. It is also diagonal and full-rank.

2.5 Eigenvalue Decomposition and Spectral Theorem

- Consider $A \in \mathbb{R}^{m\times n}$ with eigenvalues $\lambda_1, \ldots, \lambda_n$. Let $u^{(1)}, \ldots, u^{(n)}$ be arbitrary eigenvectors each associated with one eigenvalue.
- Assume $u^{(1)}, \ldots, u^{(n)}$ are linearly independent. Then, A can be decomposed as $U\Lambda U^{-1}$,
  where $U = \begin{bmatrix} u^{(1)} & \ldots & u^{(n)} \end{bmatrix}$ and $\Lambda = \text{diag}(\lambda_1, \ldots, \lambda_n)$. We say A is a diagonalizable matrix.
- If $\lambda_1, \ldots, \lambda_n$ are all distinct, A is always diagonalizable. If A has repeated eigenvalues, Theorem 3.4 of our textbook *Optimization Model. G.C. Calafiore and L. El Ghaoui* explains when linearly independent eigenvectors exist.
- Spectral theorem: Consider a *symmetric* matrix $A \in \mathbb{S}^n$. For each eigenvalue λ_i, select an eigenvector $u^{(i)}$ with length 1 to assemble the matrix U. Then, it holds that $A = U\Lambda U^\top$, i.e., U is an orthogonal matrix.
- Symmetric matrices are always diagonalizable.

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2.6 Singular Value Decomposition (SVD)

- SVD Theorem: Given an arbitrary (not necessarily square) matrix $A \in \mathbb{R}^{m\times n}$, there exist matrices $U \in \mathbb{R}^{m\times m}$, $V \in \mathbb{R}^{n\times n}$, and $\Sigma \in \mathbb{R}^{m\times n}$ such that:

  1. $A = U\Sigma V^\top$.
  2. U and V are each orthogonal matrices, i.e., $U^\top U = I_m$ and $V^\top V = I_n$.
  3. Σ is a "rectangular diagonal matrix" in the form of $\begin{bmatrix} \sigma_1 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_m & 0 & \cdots & 0 \end{bmatrix}$ if $n \geq m$

     and in the form of $\begin{bmatrix} \sigma_1 & 0 & \cdots & 0 \\ 0 & \sigma_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_n \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}$ if $n \leq m$, where $\sigma_1 \geq \sigma_2 \geq \cdots \geq 0$.

- $\sigma_1, \sigma_2, \ldots$ are called the singular values of A.

- Let r be the number of non-zero singular values of A, i.e., $\underbrace{\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_r}_{\text{non-zero}} > \sigma_{r+1} = \sigma_{r+2} = \ldots = 0$. It holds that $r = \text{Rank}(A)$.

- If A is symmetric and PSD, then its eigenvalues and singular values are the same, and its eigenvalue decomposition $A = U\Lambda U^\top$ is a valid SVD. However, eigenvalues and singular values are different in general.

- Finding SVD by hand:
  The non-zero singular values of A are the square root of the non-zero eigenvalues of $AA^\top$ or $A^\top A$.
  The columns of U (called the left singular vectors) are the eigenvectors of $AA^\top$.
  The columns of V (called the right singular vectors) are the eigenvectors of $A^\top A$.

- If αA, where α is some non-negative real scalar, is an orthogonal matrix, then one possible SVD for A is $A = I_n \frac{I_n}{\alpha}(\alpha A)$.

2.7 Matrix Pseudo-Inverse

- The pseudo-inverse (or Moore-Penrose inverse) of a matrix $A = U\Sigma V^\top$ is $A^\dagger = V\Sigma^\dagger U^\top$, where

$$\Sigma^\dagger = \begin{bmatrix} 1/\sigma_1 & \cdots & 0 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 1/\sigma_r & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \cdots & 0 & 0 & \cdots & 0 \end{bmatrix} \text{ if } n \leq m \text{ and } \Sigma^\dagger = \begin{bmatrix} 1/\sigma_1 & \cdots & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \cdots & 1/\sigma_r & \cdots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \cdots & 0 \\ 0 & \cdots & 0 & \cdots & 0 \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ 0 & \cdots & 0 & \cdots & 0 \end{bmatrix} \text{ if } n \geq m,$$

  i.e., we take the inverse of the positive singular values and fill the rest with zero.

- If A is invertible, then $A^\dagger = A^{-1}$ and therefore $AA^\dagger = I_n$. However, $AA^\dagger$ does not produce I_n in general.

- If $A \in \mathbb{R}^{m\times n}$ has linearly independent rows, i.e., $n \geq m = \text{Rank}(A)$, then $A^\dagger = A^\top(AA^\top)^{-1}$.
  If $A \in \mathbb{R}^{m\times n}$ has linearly independent columns, i.e., $m \geq n = \text{Rank}(A)$, then $A^\dagger = (A^\top A)^{-1}A^\top$.

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2.8 Matrix Norms

Consider a matrix $A \in \mathbb{R}^{m \times n}$.

- Frobenius norm: $\|A\|_F := \|\text{vec}(A)\|_2$, where the vector $\text{vec}(A) \in \mathbb{R}^{mn}$ is a concatenation of all columns of A. I.e., `A_frob = (A ** 2).sum().sqrt()` with Python-like pseudo code.
- It holds that $\|A\|_F^2$ is equal to the sum of the squared singular values of A, i.e., $\|A\|_F^2 = \sum_{i=1}^r \sigma_i^2(A)$.
- ℓ_p-induced norm: $\|A\|_p := \max_{z \in \mathbb{R}^n, z \neq 0} \frac{\|Az\|_p}{\|z\|_p} = \max_{\|w\|_p = 1} \|Aw\|_p$.
- One example of the ℓ_p-induced norm is the *spectral norm* for $p = 2$.
  - It holds that $\|A\|_2 = \sigma_1(A) = \sqrt{\lambda_{\max}(A^\top A)}$, where $\sigma_1(A)$ is the largest singular value of A and $\lambda_{\max}(A^\top A)$ is the largest eigenvalue of $A^\top A$.

3 Set Theory

3.1 Basic Set Theory

- A set $\mathcal{S} \subseteq \mathbb{R}^n$ is open if for every $x \in S$, there exists $\epsilon > 0$ such that $B_\epsilon(x) \subset S$, where $B_\epsilon(x)$ is a ball centered at x with radius ϵ.
- A set $\mathcal{S} \subseteq \mathbb{R}^n$ is closed if its complement $\mathbb{R}^n \setminus S$ is open.
- A set $\mathcal{S} \subseteq \mathbb{R}^n$ is bounded if there exists an $r > 0$ such that $S \subseteq B_r(0)$.
- A set is compact if it is closed and bounded.
- Given a set $\mathcal{S} \subseteq \mathbb{R}^n$, a point $x \in \mathcal{S}$ is said to belong to the *interior* of $\mathcal{S}$ is we can draw a ball in $\mathbb{R}^n$ centered at x and of non-zero radius that belongs to $\mathcal{S}$. The interior is denoted as $\text{int}\,\mathcal{S}$.
- The closure of a set is defined as:

$$\text{cls}(\mathcal{S}) = \left\{ z \in \mathbb{R}^n \;\middle|\; z = \lim_{k \to \infty} x^{(k)} \text{ where } x^{(k)} \in \mathcal{S},\ \forall k \right\}$$

- Boundary of a set is $\partial \mathcal{S} = \text{cls}(\mathcal{S}) \setminus \text{int}(\mathcal{S})$.
- Given a convex set $\mathcal{S} \subseteq \mathbb{R}^n$, a point $x \in \mathcal{S}$ is said to belong to the *relative interior* of $\mathcal{S}$ is we can draw a ball in the affine hull of $\mathcal{S}$ centered at x and of non-zero radius that belongs to $\mathcal{S}$. The relative interior is denoted as $\text{relint}\,\mathcal{S}$.

3.2 Affine Sets, Convex Sets, and Dimension

- Affine combination of $x_1, \ldots, x_k \in \mathbb{R}^n$: $\left\{ \sum_{i=1}^k \alpha_i x_i \;\middle|\; \sum_{i=1}^k \alpha_i = 1 \right\}$.
- Convex combination $x_1, \ldots, x_k \in \mathbb{R}^n$: $\left\{ \sum_{i=1}^k \alpha_i x_i \;\middle|\; \sum_{i=1}^k \alpha_i = 1,\ \alpha_i \geq 0,\ \forall i \right\}$.
- A set $\mathcal{S}$ is affine if for $x, y \in \mathcal{S}$ and $t \in \mathbb{R}$, the affine combination $tx + (1-t)y$ is in $\mathcal{S}$ (also recall that Section 1.1 introduced affine sets based on subspaces).
  - A hyperplane is an affine set, but a half-space is not.
- A set $\mathcal{S}$ is convex if for $x, y \in \mathcal{S}$ and $t \in [0, 1]$, the convex combination $tx + (1-t)y$ is in $\mathcal{S}$.
  - A polyhedron $\left\{ x \;\middle|\; a_i^\top x \leq b_i,\ c_j^\top x = d,\ \forall i, j \right\}$ is a convex set.
  - Norm balls and half-spaces are convex.

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- The set of PD matrices is convex, and the set of PSD matrices is also convex.

- The affine hull of a set is the smallest affine set containing the set. It is the set of affine combinations of any k points in the set.
- The convex hull of a set is the smallest convex set containing the set. It is the set of convex combinations of any k points in the set.
- Operations that preserve set convexity:
  - The intersection of convex sets is convex (note that the union of convex sets may not be convex).
  - Affine transformation: consider a convex set $\mathcal{S}$ and an affine function $f : \mathbb{R}^n \to \mathbb{R}^m$. The set $\bar{\mathcal{S}} := \{f(x) : x \in \mathcal{S}\}$ is convex.
  - Projections of a convex set are convex.
- Dimension of a set $S \subseteq \mathbb{R}^n$:
  1. If S is a subspace, dimension is minimum number of spanning vectors.
  2. If S is affine, then $S = x_0 + V$, where V is a subspace, and the dimension of S is the dimension of V.
  3. If S is convex, dimension of S is defined as the dimension of the affine hull of S.

3.3 Separating Hyperplane

- Supporting hyperplane theorem: for a convex set C and a boundary point $z \in \partial C$, we can always find a *supporting hyperplane* $H = \{x \in \mathbb{R}^n \mid a^\top x = b\}$ which is defined as satisfying:
  1. $z \in H$
  2. $C \subseteq H_-$, where $H_- = \{x \in \mathbb{R}^n \mid a^\top x \leq b\}$
- Separating hyperplane: for two convex sets $C_1, C_2 \subseteq \mathbb{R}^n$, a hyperplane $H = \{x \in \mathbb{R}^n \mid a^\top x = b\}$ is said to separate C_1 and C_2 if
  1. $C_1 \subseteq H_-$, where $H_- = \{x \mid a^\top x \leq b\}$,
  2. $C_2 \subseteq H_+$, where $H_- = \{x \mid a^\top x \geq b\}$.

  If $H \cap C_1 = H \cap C_2 = \emptyset$, then H *strictly separates* C_1 and C_2.
- Separating hyperplane theorem: assume C_1, C_2 are convex. Then we have two statements
  1. if $C_1 \cap C_2 = \emptyset$, then a separating hyperplane exists.
  2. if $C_1 \cap C_2 = \emptyset$, C_1 and C_2 are closed, and either C_1 or C_2 are bounded, then a strictly separating hyperplane exists.

4 Optimization Problems

4.1 Standard Form and Basic Properties

- Consider functions $f_i : \mathbb{R}^n \mapsto \mathbb{R}$ for $i = 0, \ldots, n$. The standard form of optimization problems is

$$\min_{x \in \mathbb{R}^n} \quad f_0(x) \quad \text{subject to} \quad f_i(x) \leq 0, \ \ \forall i = 1, \ldots, m. \tag{1}$$

- Equality constraints can be converted into inequality constraints.
  For some function $h : \mathbb{R}^n \mapsto \mathbb{R}$, it holds that $h(x) = 0 \iff \{h(x) \leq 0, -h(x) \leq 0\}$.
- Consider the optimization problem (1). A point $y \in \mathbb{R}^n$ is called feasible if $f_i(y) \leq 0$ for all $i \in 1, \ldots, m$.
  Furthermore, the feasible set $\mathcal{X}$ is the set of all feasible points: $\mathcal{X} = \{x \in \mathbb{R}^n \mid f_i(x) \leq 0, \ \forall i \in 1, \ldots, m\}$.
- A point $x^\star \in \mathbb{R}^n$ is a global minimum if $f_0(x^\star) \leq f_0(x)$ for all $x \in \mathcal{X}$.
- Consider an arbitrary function $f(x)$. Suppose that some x is the optimal solution to $\min_x f(x)$, then it is also optimal for $\max_x -f(x)$ and $\min_x \alpha f(x)$, where $\alpha > 0$ is any positive scalar.

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4.2 Optimization Problem Solution Types

- Infeasible: There is no input that satisfies all the constraints. E.g., we have constraints $x > 1$ and $x < 0$.
- Unbounded: The optimal objective value of the minimization problem is $-\infty$. E.g., minimize x without constraints.
- Unattainable: There is no finite solution. E.g., minimize $\frac{1}{x}$ subject to $x > 0$ (we can always improve the solution by increasing x).
- Tractable: There is an algorithm to solve it efficiently (polynomial time). Otherwise, the problem is intractable.
- For minimization problems, the optimal objective value is $+\infty$ if infeasible, $-\infty$ if unbounded from below, and finite otherwise ($x^\star$ may or may not be attainable). Maximization problems see the opposite.

4.3 Coercive Functions and Finite Solutions

- Optimization problems might not have a finite solution (e.g., $\min_x e^x$).
- Need a definition: a function $f : \mathbb{R}^n \to \mathbb{R}$ is *coercive* if $\lim_{\|x\|\to\infty} f(x) = \infty$.
  - Note that $f(x)$ must tend to $+\infty$ along all directions when $\|x\| \to \infty$ to be coercive.
  - Conversely, to prove a function is not coercive, we just need to find one direction along which $f(x)$ does not go to $+\infty$ when $\|x\| \to \infty$.
- Theorem (unconstrained): consider $f : \mathbb{R}^n \to \mathbb{R}$ with domain $\mathbb{R}^n$ (either convex or non-convex). Then if f is continuous and coercive, $\min f(x)$ has a finite solution.
- Theorem (constrained):
  1. Consider $\min f(x)$ subject to $x \in S$.
     Suppose that f (convex or nonconvex but with domain $\mathbb{R}^n$) is coercive and continuous.
     If S (convex or nonconvex) is closed, the optimization problem has a finite solution.
  2. Consider the optimization problem

$$\begin{aligned} \min \quad f_0(x) \quad \text{subject to} \quad & f_i(x) \le 0 \qquad i = 1, \dots, m \\ & h_j(x) = 0 \qquad j = 1, \dots, k, \end{aligned} \tag{2}$$

     where f_0, f_i's, and h_j's are arbitrary but continuous with domain $\mathbb{R}^n$. Then if f_0 is coercive, the optimization has a finite solution.
- What if we don't have coerciveness? Then there is an alternative theorem.
- Weierstrass theorem: consider $\min f(x)$ s.t. $x \in S$, where $f : \mathbb{R}^n \to \mathbb{R}$ is continuous. If S is compact, then the optimization has a finite solution.
  - So for an optimization of the form (2), as long as f_0 is continuous and the feasible set is bounded, we have a finite solution.

4.4 Convex Functions

- A function $f : \mathbb{R}^n \to \mathbb{R}$ is convex if and only if its domain is a convex set and $f(\alpha x + (1-\alpha)y) \le \alpha f(x) + (1-\alpha) f(y)$.
  - This is the zeroth-order condition for convexity.
  - Replacing $\le$ with $<$ gives the definition of strict convexity.
  - Thus, the set defined by $\{x : f(x) \le 0\}$, where f is a convex function, is a convex set.
- A function is called concave if its negative is convex.

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- Convexity does not imply continuity.
  - Example: consider an end point $\tilde{x}$ of dom$f$. f can still be convex if it "jumps up" at $\tilde{x}$.
  - The discontinuity should happen only on the boundaries.
- First-order convexity condition: $f(y) + \nabla f(y)^\top (x - y) \leq f(x)$ for all $x, y \in$ dom$f$ (replace with $<$ for strict convexity).
- Second-order convexity condition: f is convex if and only if $\nabla^2 f(x) \succeq 0$ for all $x \in$ dom$f$.
  - If $\nabla^2 f(x) \succ 0$ for all $x \in$ dom$f$, then f is strictly convex.
    The reverse direction may not hold (e.g., $f(x) = x^4$).
- Some example convex functions:
  - $f(x) = e^{ax}$.
  - $f(x) = x^a$ where $a \geq 1$ or $a \leq 0$ on $\mathbb{R}_{++}$.
  - $f(x) = -\log(x)$ on $\mathbb{R}_{++}$.
  - Any ℓ_p form function $f(x) = \|x\|_p$.
  - Quadratic functions $f(x) = x^\top P x + q^\top x + r$, where P is symmetric and $P \succeq 0$. If $P \succ 0$, f is strictly convex.
- Some properties of convex functions:
  - Affine functions are simultaneously convex and concave.
  - The point-wise maximum of a set of convex functions is convex.
    The point-wise minimum of a set of concave functions is concave.
  - A summation of convex functions $f(x) := \sum_{i=1}^{k} \alpha_i f_i(x)$ for $a_i \geq 0$ is convex if f_i is convex for all i.
  - If $f(x)$ is convex, then the affine transformation $g(x) = f(Ax + b)$ is also convex.
  - If $f(x)$ is convex and $g(x)$ is convex and non-decreasing, then the composite function $g \circ f(x)$ is convex.
  - **Compositions of convex functions are not convex in general.**

4.5 Convex Optimization Problems

- Consider an optimization problem $\min_x f(x)$ subject to $x \in \mathcal{X}$. This problem is convex when
  - f is a convex function (see Section 4.4);
  - $\mathcal{X}$ is a convex set (see Section 3.2).
- Consider an optimization problem $\min_x f(x)$ subject to $g_i(x) \leq 0$ for all i and $h_j(x) = 0$ for all j.
  This problem is convex when
  - f is a convex function;
  - g_i is a convex function for each i;
  - h_j is an *affine* function for each j.
- For a convex optimization problem:
  - All local solutions are global.
  - The feasible set is a convex set.
  - The set of all global minima is a convex set.
  - If the objective is strictly convex, then there is either no solution or a unique solution.

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4.6 Linear Programming (LP)

- Linear program can be written as:

$$\min_x \quad a_0^\top x \quad \text{subject to} \quad \begin{aligned} a_i^\top x - b_i &= 0 \quad i = 1, \ldots, m, \\ c_j^\top x - d_j &\leq 0 \quad j = 1, \ldots, k. \end{aligned}$$

- Rewritten in matrix form:

$$\min_x \quad a_0^\top x \quad \text{subject to} \quad Ax = b, \quad Cx \leq d. \tag{3}$$

- Rewritten in standard form:

$$\min_x \quad a_0^\top x \quad \text{subject to} \quad Ax = b, \quad x \geq 0. \tag{4}$$

  - If an LP is reformulated from the form (3) into the standard form (4), A and b in the standard form can be different from the ones the original form.
  - To convert an affine inequality constraint $Cx \leq d$ into the standard form, we can introduce a slack variable s (same shape as d) and rewrite the constraint as $Cx + s = d$ and $s \geq 0$.
  - The constraint $x \geq 0$ must apply to all optimization variables. If some variables are not constrained to be non-negative in the original problem (say x_i is one of such variables), we can "split" it into $x_{i+} \geq 0$ and $x_{i-} \geq 0$ and represent x_i as $x_{i+} - x_{i-}$.

- Algorithms to solve LPs:
  - Simplex: start at an arbitrary vertex and repeatedly go to a neighbor vertex with lower objective value.
  - Interior point: start in the interior of polyhedron and move towards optimal solution (stays in the interior as opposed to moving on the boundary).

4.6.1 LP Solutions are at Feasible Set Vertices

- Definition: for a convex set S, a point $y \in S$ is an *extreme point* if there do not exists points $u, v \in S$ such that $y = \alpha u + (1 - \alpha)v$ for some $0 < \alpha < 1$.
  - Extreme points of a polyhedron are called *vertices*.
- Theorem: assume LP has a solution. Then one of its feasible set vertices is *a* solution (could have other solutions as well).
- Theorem: if an LP's feasible set is bounded, then a solution exists.

4.6.2 Finding All Vertices

Consider a feasible set for x defined by $Ax = b$ and $x \geq 0$, where $x \in \mathbb{R}^n$, $A \in \mathbb{R}^{m \times n}$, and $b \in \mathbb{R}^m$. Assume that $m \leq n$, i.e., A is wide.

An algorithm that finds all vertices of the feasible set is as follows.

- Find all possible combinations of m columns of A and denote the resulting square sub-matrices formed by these columns as $A_i^{\text{sub}} \in \mathbb{R}^{m \times m}$ for $i = 1, \ldots, \binom{n}{m}$, where $\binom{n}{m}$ denotes n-choose-m and is equal to $\frac{n!}{m!(n-m)!}$.
- Then, the number of vertices is the number of A_i^{sub} matrices that satisfies
  1. A_i^{sub} is invertible;
  2. The solution $z^\star$ to the linear system $A_i^{\text{sub}} z = b$ is feasible (i.e., non-negative).

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4.6.3 Converting to LP via Epigraph Formulation

Sometimes, can convert an optimization problem into an LP via an epigraph formulation.

- Start with general optimization problem where $\mathcal{S}$ is some feasible set:

$$\min_{x \in \mathbb{R}^n} \quad f(x) \quad \text{subject to} \quad x \in \mathcal{S}.$$

- Reformulate using a slack variable t:

$$\min_{x \in \mathbb{R}^n, t \in \mathbb{R}} t \quad \text{subject to} \quad x \in \mathcal{S}, \quad f(x) \leq t.$$

- For example, $\min_{x \in \mathbb{R}^n} \|x\|_\infty$ subject to $x \in \mathcal{S}$
  can be converted to $\min_{x \in \mathbb{R}^n, t \in \mathbb{R}} t$ subject to $x \in \mathcal{S}$ and $\|x\|_\infty \leq t$,
  where $\|x\|_\infty \leq t \iff |x_i| \leq t$ for all i.

4.7 Quadratic Programming (QP)

- QP includes a quadratic term in the objective, where $P_0 \succeq 0$:

$$\min_{x \in \mathbb{R}^n} \quad x^\top P_0 x + q_0^\top x + r_0 \quad \text{subject to} \quad Ax = b, \quad Cx \leq d.$$

4.8 Quadratically Constrained Quadratic Program (QCQP)

- A QCQP can be written in the form of

$$\begin{aligned} \min_{x \in \mathbb{R}^n} \quad x^\top P_0 x + q_0^\top x + r_0 \quad \text{subject to} \quad & Ax = b, \\ & x^\top P_j x + q_j^\top x + r_j \leq 0 \ \text{ for } \ j = 1, \ldots, k, \end{aligned}$$

  where $P_j \succeq 0$ for $j = 0, \ldots, k$.

- Hierarchy of optimization problems: LP $\subset$ QP $\subset$ QCQP $\subset$ convex optimization.

4.9 Convex Relaxations

- Consider optimization problem with $f(x)$ convex but S non-convex:

$$\min_{x \in \mathbb{R}^n} \quad f(x) \quad \text{subject to} \quad x \in S$$

- If we replace S with a convex $\tilde{S}$ such that $S \subset \tilde{S}$, we get a *convex relaxation*:

$$\min_{x \in \mathbb{R}^n} \quad f(x) \quad \text{subject to} \quad x \in \tilde{S}$$

- Let x^* and $\tilde{x}$ be global minima of the original and relaxed optimizations, respectively. Then
  - $f(\tilde{x}) \leq f(x^*)$
  - If $\tilde{x} \in S$, then $\tilde{x}$ is a global min for original optimization problem.

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4.10 Integer Programming (IP)

- An IP is just an LP with a constraint that all elements of x are integers:

$$\min_x \quad a_0^\top x \quad \text{subject to} \quad \begin{aligned} & Ax = b, \\ & x \geq 0, \\ & x_i \ \text{ are integers for } \ i = 1, \ldots, n. \end{aligned}$$

- IPs are non-convex!
- *Can form a convex relaxation by dropping the integer constraint.*
  - Let P_1 be the above IP, and let P_2 be the corresponding relaxed LP dropping the integer constraint.
  - Theorem: if all vertices of the feasible set of P_2 are integral, then the convex relaxation is exact, and the optimal objectives of P_1 and P_2 are equal.
  - This is the case for assignment / transport problems (see Lecture 19)!

5 Optimality Conditions

5.1 Gradient and Hessian

Consider a function $f(x) : \mathbb{R}^n \mapsto \mathbb{R}$ and assume $f(x)$ is twice continuously differentiable. Let x_i denote the i-th entry of x for $i = 1, \ldots, n$.

- The gradient is an n-dimensional vector $\nabla f(x) := \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \ldots, \frac{\partial f}{\partial x_n}\right)$.
- The Hessian is an $n \times n$ symmetric matrix $\nabla^2 f(x) := \begin{bmatrix} \frac{\partial^2 f}{\partial x_1 \partial x_1} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n \partial x_n} \end{bmatrix}$.
- If $n = 1$, then the gradient is the first-order derivative and the Hessian is the second-order derivative.
- Suppose that $f(x)$ is quadratic, i.e., $f(x) = x^\top P x + q^\top x + r$ for some $P \in \mathbb{S}^n$, $q \in \mathbb{R}^n$, and $r \in \mathbb{R}$. Then, it holds that $\nabla f(x) = 2Px + q$ and $\nabla^2 f(x) = 2P$.
- Gradient chain rule: Consider functions $f : \mathbb{R}^m \to \mathbb{R}$ and $g : \mathbb{R}^n \to \mathbb{R}^m$. Define $\phi(x) := f(g(x))$. Then

$$\underbrace{\nabla \phi(x)}_{n\text{-dimensional vector}} = \underbrace{\begin{bmatrix} \nabla g_1(x) & \ldots & \nabla g_m(x) \end{bmatrix}}_{n \times m \text{ matrix}} \times \underbrace{\nabla f(z)|_{z=g(x)}}_{m\text{-dimensional vector}}.$$

- Taylor series approximation: given a function $f(x) : \mathbb{R}^n \to \mathbb{R}$ that is differentiable at $x_0 \in \mathbb{R}^n$, it can be approximated by an affine function in a neighborhood of x_0:

$$f(x) = f(x_0) + \nabla f(x_0)^\top (x - x_0) + \epsilon(x),$$

  where $\epsilon(x)$ goes to zero faster than first order, i.e., $\lim_{x \to x_0} \frac{\epsilon(x)}{\|x - x_0\|} = 0$.
  - So, to the first order we have the approximation: $f(x) \approx f(x_0) + \nabla f(x_0)^\top (x - x_0)$.

5.2 Optimality Conditions for Unconstrained Optimization Problems

Consider the optimization problem $\min_{x \in \mathbb{R}^n} f(x)$, where f is differentiable.

- First-order necessary condition: If $x^\star$ is a local minimum, then $\nabla f(x^\star) = 0$.
- Suppose that $\nabla^2 f(x) \succeq 0$ for *all* $x \in \mathbb{R}^n$, i.e., the problem is convex. Then,
  - All local minima are global minima.
  - $x^\star$ is a global minimum (and a local minimum) if and only if $\nabla f(x^\star) = 0$.

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5.3 Slater's Condition

Slater's condition is a widely used regularity condition.

Consider a convex problem $\min_x f_0(x)$ subject to $f_i(x) \leq 0$ for $i = 1, \ldots, k$ and $h_j(x) = 0$ for $j = 1, \ldots, m$. Denote the intersection of each f_i's domain as $\mathcal{D}$.

- Slater's condition holds if there exists a point $y \in \operatorname{relint} \mathcal{D}$ such that
  - $h_j(y) = 0$ for $j = 1, \ldots, m$.
  - $f_i(y) \leq 0$ for all affine f_i.
  - $f_i(y) < 0$ for all non-affine f_i.
- y is not unique in general.
- When there are no constraints, Slater's condition holds by convention.
- When all constraints are affine, e.g., LP or QP, Slater's condition is equivalent to feasibility.

5.4 Optimality Conditions for Constrained Optimization Problems

Consider the optimization problem

$$\begin{aligned} \min_x f_0(x) \quad \text{subject to} \quad & f_i(x) = 0,\ i = 1, \ldots, k, \\ & h_j(x) \leq 0,\ j = 1, \ldots, m. \end{aligned}$$

Denote the dual variables associated with the equality constraints as $\mu_1, \ldots, \mu_k$. Similarly, denote the dual variables associated with the inequality constraints as $\lambda_1, \ldots, \lambda_m$. The Lagrangian of this problem is then

$$L(x, \lambda, \mu) \coloneqq f(x) + \sum_{i=1}^{k} \mu_i f_i(x) + \sum_{j=1}^{m} \lambda_j h_j(x).$$

- **First-Order Necessary Condition (FOC):** If $x_\star$ is a regular point and a local minimum, then there exist Lagrangian multipliers $\lambda_{1\star}, \ldots, \lambda_{m\star}$ and $\mu_{1\star}, \ldots, \mu_{l\star}$ that makes the following holds true:
  1. **Primal Feasibility**: $f_i(x_\star) = 0$ for all $i = 1, \ldots, k$ and $h_j(x_\star) \leq 0$ for all $j = 1, \ldots, m$;
  2. **Dual Feasibility**: $\lambda_{j\star} \geq 0$ for all $j = 1, \ldots m$;
  3. **Lagrangian Stationarity**: $\nabla f(x_\star) + \sum_{i=1}^{k} \mu i\star \nabla f_i(x_\star) + \sum_{j=1}^{m} \lambda_{j\star} \nabla h_j(x_\star) = 0$;
  4. **Complementary Slackness**: $\lambda_{j\star} \cdot h_j(x_\star) = 0$ for all $j = 1, \ldots m$.

  These conditions are called the Karush–Kuhn–Tucker (**KKT**) conditions.
- For convex optimization problems, the KKT conditions are sufficient can be used to find global optima.
- Consider a quadratic optimization with equality constraints in the form of $\min_{x \in \mathbb{R}^n} x^\top P_0 x + q_0^\top x + r_0$ subject to $Ax = b$ where $P_0 \in \mathbb{S}_+^n$, $A \in \mathbb{R}^{m \times n}$, $q_0 \in \mathbb{R}^n$, and $r_0 \in \mathbb{R}$.

  Suppose that Slater's condition holds, i.e., $Ax = b$ admits one or more solutions. Then, the optimal primal-dual solution $(x^\star, \mu^\star)$ satisfies $\begin{bmatrix} A & \mathbf{0}_{m \times m} \\ 2P_0 & A^\top \end{bmatrix} \begin{bmatrix} x^\star \\ \mu^\star \end{bmatrix} = \begin{bmatrix} b \\ -q_0 \end{bmatrix}$.

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6 Linear Systems, Least Squares, and Regression

6.1 Solving Linear Systems

Consider solving a system of linear equations $Ax = y$.

- $Ax = y$ has a unique solution if and only if $y \in \mathcal{R}(A)$ and $\mathcal{N}(A) = \{0\}$.
- If A's nullspace satisfies $\mathcal{N}(A) \neq \{0\}$, any solution $x^\star$ produces a space of solutions $x^\star + z$ where $z \in \mathcal{N}(A)$.
- Tall matrix: if $A \in \mathbb{R}^{m\times n}$, where $m > n$, then we have an overdetermined case, and there is likely no solution unless we are lucky and $y \in \mathcal{R}(A)$.
- Fat matrix: now assume $n > m$, and our rows are linearly independent. Now we have an underdetermined case, and the solution space is $\bar{x} + \mathcal{N}(A)$ where $\bar{x}$ is an arbitrary solution.
  For many applications, the "best" solution is the one with minimum norm:
  $$\min_{x\in\mathbb{R}^n} \|x\| \quad \text{subject to} \quad Ax = y.$$
  The minimum-norm solution can be derived as $x^\star = A^\top(AA^\top)^{-1}y = A^\dagger y$.
- If A is square and full-rank (invertible), we can solve directly $x = A^{-1}y$.

6.2 Least Squares (LS)

What if we are in the overdetermined case and y is not in the range of A? We need to minimize how much we violate the equation $Ax = y$, instead of solving it exactly.

- Given a matrix $A \in \mathbb{R}^{m\times n}$ and a vector $y \in \mathbb{R}^m$, we aim to solve the problem $\min_{x\in\mathbb{R}^n} \|Ax - y\|_2$.
- Denote the optimal solution as $x^\star$. Note that $x^\star$ also solves $\min_{x\in\mathbb{R}^n} \|Ax - y\|_2^2$.
- The set of solutions for the LS problem is $\mathcal{S} := \{x^\star \mid A^\top Ax^\star = A^\top y\}$. Proof: optimality conditions.
- It holds that $\mathcal{S} = A^\dagger y + \mathcal{N}(A)$, where $A^\dagger$ is the pseudo-inverse of A as defined above.

6.3 Relationships between Least Squares and Projection

- Geometrically, the LS problem finds the projection of y onto $\mathcal{R}(A)$, the range of A.
- The projection result $y^\star := Ax^\star = \Pi_{\mathcal{R}(A)}y$ exists and is unique.
- Theorem on projection: $y - y^\star \perp \mathcal{R}(A)$. I.e., $\langle y - y^\star, v\rangle = 0$ for all $v \in \mathcal{R}(A)$.
- We can find $y^\star$ by solving for the vector that simultaneously satisfies $y^\star \in \mathcal{R}(A)$ and $y - y^\star \perp \mathcal{R}(A)$.

6.4 Minimum-Norm Solution to Least Squares

- To find the minimum-norm solution, solve $\min_{x\in\mathcal{S}} \|x\|_2$. I.e., $\min_{x\in\mathbb{R}^n} \|x\|_2$ subject to $A^\top Ax = A^\top y$.
- The minimum-norm LS solution is unique and equal to $A^\dagger y = (A^\top A)^{-1}A^\top y$.
- If A has full column rank, i.e., $m \geq n = \text{Rank}(A)$, then $A^\top A$ is invertible and $\mathcal{N}(A) = \{0\}$.
  In this case, $x^\star = A^\dagger y$ is the unique LS solution.

6.5 Ridge Regression

- An ℓ_2-regularized LS problem: $\min_{x\in\mathbb{R}^n} \|Ax - y\|_2^2 + \alpha\|x\|_2^2$ where α is a non-negative scalar.
- The matrix $A^\top A + \alpha I_n$ is invertible, and the unique solution to the ridge regression problem is $x^\star = (A^\top A + \alpha I_n)^{-1}A^\top y$.

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6.6 Sparsity and LASSO Regression

- $x \in \mathbb{R}^n$ is called sparse if many of its entries are zero. Otherwise it is called dense.
- The number of non-zero entries of x is called its *cardinality*, denoted as $\|x\|_0$. When all entries of x are within $[-1, 1]$, it holds that $\|x\|_1 \leq \|x\|_0$.
- LASSO is an ℓ_1-regularized LS problem that promotes solution sparsity:

$$\min_{x \in \mathbb{R}^n} \|Ax - y\|_2^2 + \alpha \|x\|_1,$$

  where α is a non-negative scalar.
- LASSO's objective function is not always differentiable. However, it can be reformulated as a QP via the epigraph method:

$$\min_{x,t \in \mathbb{R}^n} x^\top P_0 x + q_0^\top x + r_0 + \alpha \sum_{i=1}^n t_i \quad \text{subject to} \quad -t_i \leq x_i \leq t_i \;\; \text{for} \;\; i = 1, \ldots, n,$$

  where $P_0 \in \mathbb{S}_+^n$, $q_0 \in \mathbb{R}^n$, and $r_0 \in \mathbb{R}$ are expressions of A and y.
- $x^\star$ is a solution to LASSO if and only if $2P_0 x^\star + q_0 + \lambda^\star = 0$, where each entry of $\lambda^\star$ satisfies:
  $\lambda_i^\star = \alpha$ if $x_i^\star \geq 0$, $\lambda_i^\star = -\alpha$ if $x_i^\star \leq 0$, and $\lambda_i^\star \in [-\alpha, \alpha]$ if $x_i^\star = 0$.
  Furthermore, it holds that $|x_i^\star| = t_i^\star$ for all i.

6.7 Sensitivity Analysis – Linear Systems

- Consider system of linear equations with $A \in \mathbb{R}^{n \times n}$ invertible and $y \in \mathbb{R}^n$ given; we want to find $x : Ax = y$.
- Due to invertibility solution is given by $A^{-1}y$.
- What if y changes to $y + \Delta y$ due to measurement noise?
- Consider solution change to $x + \Delta x$:

$$A(x + \Delta x) = y + \Delta y \quad \text{and} \quad Ax = y \quad \Longrightarrow \quad \Delta x = A^{-1} \Delta y.$$

- Lemma: for matrix B and vector y: $\|By\|_2 \leq \|B\|_2 \|y\|_2$
- So we have $\|\Delta x\|_2 \leq \|A^{-1}\|_2 \|\Delta y\|_2$ and $\|y\|_2 \leq \|A\|_2 \|x\|_2$.
- Combining these two yields that

$$\frac{\|\Delta x\|_2}{\|x\|_2} \leq \|A\|_2 \|A^{-1}\|_2 \frac{\|\Delta y\|_2}{\|y\|_2}.$$

- Define condition number $\kappa(A) = \|A\|_2 \|A^{-1}\|_2$.
- Theorem: the relative change in x with regard to a relative change in y, when solving $y = Ax$ for A invertible, is given by

$$\frac{\|\Delta x\|_2}{\|x\|_2} \leq \kappa(A) \frac{\|\Delta y\|_2}{\|y\|_2}.$$

- Recall that $\|A\|_2 = \sigma_1$ is the largest singular value and $\|A^{-1}\|_2 = \frac{1}{\sigma_n}$ is the largest singular value of A^{-1}.
- If $\kappa(A)$ is close to 1, then A is called well conditioned; if $\kappa(A)$ is large, then A is ill conditioned.
- Similar bound if we perturb A to $A + \Delta A$:

$$\frac{\|\Delta x\|_2}{\|x\|_2} \leq \kappa(A) \frac{\|\Delta A\|_2}{\|A\|_2}.$$

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6.8 Sensitivity Analysis – Least Squares

- Let's consider least-square problem $\min_x \|Ax - y\|_2$, where y is a measurement vector with noise.
- How does perturbing y to $y + \Delta y$ affect solutions?
- Recall we can define an ellipse in two equivalent forms:

$$E = \{x \in \mathbb{R}^n \mid x = By, \|y\|_2 \leq 1\};$$
$$E = \{x \in \mathbb{R}^n \mid x^\top P^{-1} x \leq 1\} \quad \text{where } P = BB^\top \text{ is PSD.}$$

  - Let $v^1, \ldots, v^n$ be eigenvectors of P with associated eigenvalues $\lambda_1, \ldots, \lambda_n$. The the ellipse has semi-axes is the directions $v^1, \ldots, v^n$ with lengths $\sqrt{\lambda_1}, \ldots, \sqrt{\lambda_n}$.
- Recall that $x^* = A^\dagger y$ solves least squares; consider $x^* + \Delta x = A^\dagger(y + \Delta y)$.
- Theorem: for an uncertainty ball on the measurement $\|\Delta y\| \leq 1$, we get an ellipsoidal uncertainty set on the solution changes:

$$E = \{\Delta x \in \mathbb{R}^n \mid \Delta x = A^\dagger \Delta y, \|\Delta y\| \leq 1\}.$$

- E is an ellipse with semi-axes $v^1, \ldots, v^n$ and lengths $\frac{1}{\sigma_1}, \ldots, \frac{1}{\sigma_r}, 0, \ldots, 0$ from the SVD $A = U\Sigma V^\top$ (this is because $A = U\Sigma V^\top \implies A^\dagger = V\Sigma^\dagger U^\top$).

7 Duality

As in Section 5.4, Consider the (potentially non-convex) optimization problem

$$\min_x f_0(x) \quad \text{subject to} \quad \begin{aligned} f_i(x) &= 0,\ i = 1, \ldots, k, \\ h_j(x) &\leq 0,\ j = 1, \ldots, m. \end{aligned}$$

Denote the global optimal solution to this problem as $x^\star$. Also denote the dual variables associated with the equality constraints as $\mu_1, \ldots, \mu_k$. Similarly, denote the dual variables associated with the inequality constraints as $\lambda_1, \ldots, \lambda_m$. Once again, the Lagrangian of this problem is

$$L(x, \lambda, \mu) \coloneqq f(x) + \sum_{i=1}^{k} \mu_i f_i(x) + \sum_{j=1}^{m} \lambda_j h_j(x).$$

7.1 Weak Duality

- In the context of duality, the original problem is called the *primal problem*. We call its optimal objective $p^\star \coloneqq f_0(x^\star)$ the *primal solution*.
- Consider arbitrary $\mu \in \mathbb{R}^k$ and $\lambda \in \mathbb{R}^m$ where $\lambda \geq 0$. It holds that $\min_x L(x, \lambda, \mu) \leq p^\star$.
- Hence, to find a meaningful lower bound to $p^\star$, we can solve

$$\max_{\mu \in \mathbb{R}^k, \lambda \in \mathbb{R}^m} \min_x L(x, \lambda, \mu) \quad \text{subject to} \quad \lambda \geq 0. \tag{5}$$

- We define $d(\lambda, \mu) \coloneqq \min_x (x, \lambda, \mu)$ as the *dual function*.

  We can then reformulate the lower bound optimization problem (5) as the maximization problem

$$d^\star \coloneqq \max_{\mu \in \mathbb{R}^k, \lambda \in \mathbb{R}^m} d(\lambda, \mu) \quad \text{subject to} \quad \lambda \geq 0, \tag{6}$$

  which we refer to as the *dual problem*. Its optimal objective $d^\star$ is called the *dual solution*.

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- It holds that $d^\star \leq p^\star$. The value of $p^\star - d^\star$ is called the *duality gap*.
- Since $d(\lambda, \mu)$ is a point-wise minimum of affine functions, it is concave no matter whether the primal problem is convex or not, and therefore (6) is always a convex optimization problem.
- Hence, leveraging weak duality, we can use convex optimization to obtain a lower bound to a hard, potentially non-convex problem.

7.2 Strong Duality

- If it holds that $p^\star = d^\star$, i.e. duality gap is zero, then *strong duality* holds.
- If the primal problem is convex and Slater's condition holds, then
  - Strong duality holds.
  - The KKT conditions of the primal problem simultaneously solve the primal problem and the dual problem. I.e., $x^\star$ solves the primal problem and $(\lambda^\star, \mu^\star)$ solves the dual problem.
  - If $x^\star$ is an arbitrary optimal solution to the primal problem and $(\lambda^\star, \mu^\star)$ is an arbitrary optimal solution to the dual problem, then $(x^\star, \lambda^\star, \mu^\star)$ satisfies the primal problem's KKT conditions.

7.3 Dual of LP and QP

- **The dual of an LP is also an LP.**

  Specifically, for some $a_0 \in \mathbb{R}^n$, $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, $C \in \mathbb{R}^{k\times n}$, and $d \in \mathbb{R}^k$, consider the LP

  $$\min_{x\in\mathbb{R}^n} \quad a_0^\top x \quad \text{subject to} \quad Ax \leq b, \quad Cx = d.$$

  The dual problem is

  $$\max_{\lambda\in\mathbb{R}^m, \mu\in\mathbb{R}^k} -\lambda^\top b - \mu^\top d \quad \text{subject to} \quad a_0 + A^\top\lambda + C^\top\mu = 0, \quad \lambda \geq 0.$$

- **The dual of a QP is also a QP.**

  Specifically, for some $P_0 \in \mathbb{S}^n_{++}$, $q_0 \in \mathbb{R}^n$, $r_0 \in \mathbb{R}$, $A \in \mathbb{R}^{m\times n}$, $b \in \mathbb{R}^m$, $C \in \mathbb{R}^{k\times n}$, $d \in \mathbb{R}^k$, consider the QP

  $$\min_{x\in\mathbb{R}^n} \quad x^\top P_0 x + q_0^\top x + r_0 \quad \text{subject to} \quad Ax \leq b, \quad Cx = d.$$

  The dual problem is

  $$\max_{\lambda\in\mathbb{R}^m, \mu\in\mathbb{R}^k} -\frac{1}{4}\left(q_0 + A^\top\lambda + C^\top\mu\right)^\top P_0^{-1}\left(q_0 + A^\top\lambda + C^\top\mu\right) - \lambda^\top b - u^\top d \quad \text{subject to} \quad \lambda \geq 0.$$

- As a special case for QP, consider the problem of finding the minimum-norm solution of a system of equations, i.e., $\min_x \|x\|_2^2$ subject to $Ax = b$ (A has full row rank). The dual problem is $\max_\mu -\frac{1}{4}\mu^\top AA^\top\mu - \mu^\top b$.

  By Lagrangian stationarity, $x^\star = -\frac{1}{2}A^\top\mu^\star$. Setting the gradient of the dual problem objective to zero gives $\mu^\star = -2(AA^\top)^{-1}b$.

7.4 Certificate and Farkas' Lemma

- Suppose that we want to show that the set $\left\{ x \in \mathbb{R}^n \;\middle|\; \begin{array}{l} f_i(x) \leq 0, \;\; i = 1, \ldots, k \\ h_j(x) = 0, \; j = 1, \ldots, m \end{array} \right\}$ is empty.

  We can consider the optimization problem $\min_x 0$ subject to $f_i(x) \leq 0$ for all i and $h_j(x) = 0$ for all j.

  Next, we find the dual function of this optimization problem $d(\lambda, \mu)$. Suppose that we can find some $(\hat{\lambda}, \hat{\mu})$ such that $d(\hat{\lambda}, \hat{\mu}) > 0$, then the optimal objective of the primal problem is $+\infty$, and hence the set of interest is empty.

- For linear case we have *Farkas' Lemma* as following.

  Equations $Ax = b$ and $x \geq 0$ have no solutions if and only if there is a solution μ to $A^\top\mu \leq 0$ and $b^\top\mu < 0$.

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8 Constraint Sensitivity Analysis

8.1 Constraint Elimination

- An inequality constraint $f_i(x) \leq 0$ is called active or binding at $x^\star$ if $f_i(x^\star) = 0$. It is called inactive or non-binding if $f_i(x^\star) < 0$.
- Suppose that $x^\star$ is an optimal solution to some problem. If we remove all non-binding solutions, then $x^\star$ is also a solution to the updated problem. However, the set of optimal solutions may change.
- Suppose that Slater's condition holds. Due to complementary slackness, the Lagrangian multipliers associated with the non-binding constraints are zero at $x^\star$.

8.2 Sensitivity Analysis

We are interested in comparing the optimization problem

$$\min_x f_0(x) \quad \text{subject to} \quad f_i(x) \leq 0 \text{ for } i = 1, \ldots, k, \quad h_j(x) = 0 \text{ for } j = 1, \ldots, m \tag{7}$$

with the problem that has perturbed constraints

$$\min_x f_0(x) \quad \text{subject to} \quad f_i(x) \leq v_i \text{ for } i = 1, \ldots, k, \quad h_j(x) = w_j \text{ for } j = 1, \ldots, m, \tag{8}$$

where each v_i and w_j is some scalar.

Denote the optimal objective value of the perturbed problem (8) as $p^\star(v, w)$. The optimal objective of the original problem (7) is $p^\star(0, 0)$. If the problem is infeasible for some (v, w), then $p^\star(v, w) = +\infty$.

We then have the following properties.

- $p^\star(v, w)$ is a convex function of v and w.
- Assume Slater's condition holds. If $p^\star(v, w)$ is differentiable at $(0, 0)$, then the Lagrangian multipliers $(\lambda^\star, \mu^\star)$ of the original problem (7) satisfies $\lambda_i^\star = -\frac{\partial p^\star(0,0)}{\partial v_i}$ for all i and $\mu_j^\star = -\frac{\partial p^\star(0,0)}{\partial w_j}$ for all j.
- As a result, it holds that $p^\star(v, w) \approx p^\star(0, 0) - \sum_i \lambda_i^\star v_i - \sum_j \mu_j^\star w_j$.

  This is the first-order Taylor's approximation for $p^\star(v, w)$. Given $x^\star$ (which can be used to compute $p^\star(0, 0)$), $\lambda^\star$, and $\mu^\star$ of the original unperturbed problem, this approximation can be computed efficiently.

  - If $\lambda_i^\star = 0$ for some i or $\mu_j^\star = 0$ for some j, then changing the corresponding constraint a little does not affect the optimal objective. Hence, those constraints can be eliminated.
  - If $\lambda_i^\star$ or $\mu_j^\star$ is small, then the optimization problem is not sensitive to the associated constraints.
  - If $\lambda_i^\star$ or $\mu_j^\star$ is large, then the optimization problem is highly sensitive to the associated constraints.

9 Numerical Optimization Algorithms

9.1 Gradient and Newton's Methods

The gradient method is a first-order method, whereas Newton's method is second-order. They apply to uni-variate and multi-variate optimization problems. Specifically, consider the problem $\min_{x \in \mathbb{R}^n} f(x)$.

- **Descent algorithm:** An iterative algorithm that generates a sequence $x^{(0)}, x^{(1)}, x^{(2)}, \ldots$ in a way that $f(x^{(k+1)}) < f(x^{(k)})$ for $k = 0, 1, 2, \ldots$.
- **Descent direction:** At a point $\bar{x} \in \mathbb{R}^n$, Δx is a descent direction if $\nabla f(\bar{x})^\top \Delta x < 0$.

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- A family of optimization algorithms can be designed with descent directions: starting from $x^{(0)}$ as the initial guess, the kth iteration is $x^{(k+1)} \leftarrow x^{(k)} + s^{(k)}\Delta x^{(k)}$ (this is called the update rule), where $\Delta x^{(k)}$ is a descent direction w.r.t. $x^{(k)}$, and $s^{(k)}$ is the step size for the kth iteration.
  - Using descent directions guarantees that $f(x^{(k+1)}) < f(x^{(k)})$ for all small enough step sizes $s^{(k)}$.
- **Gradient method:** $x^{(k+1)} \leftarrow x^{(k)} - s^{(k)}\nabla f(x^{(k)})$.
  Here, we use $-\nabla f(x^{(k)})$, which is a descent direction when $\nabla f(x^{(k)}) \neq 0$, as $\Delta x^{(k)}$.
- **Newton's method:** $x^{(k+1)} \leftarrow x^{(k)} - s^{(k)} \left(\nabla^2 f(x^{(k)})\right)^{-1} \nabla f(x^{(k)})$.
  Here, we use $-\left(\nabla^2 f(x^{(k)})\right)^{-1} \nabla f(x^{(k)})$, which is another descent direction when $\nabla f(x^{(k)}) \neq 0$ and $\nabla^2 f(x^{(k)}) \succ 0$, as $\Delta x^{(k)}$.
- If $\nabla f(x^{(k)})$ is zero, then $x^{(k)}$ is a stationary point and we stop the algorithm.
- **Why gradient/Newton?** The gradient direction minimizes a local first-order Taylor approximation of the objective function. Similarly, the Newton direction minimizes a second-order approximation, and therefore Newton's method can solve certain quadratic problems in one iteration with $s^{(k)} = 1$.
- Newton's method converges faster than the gradient method, but each iteration takes longer.
- For an iterative optimization algorithm, if $\|x^{(k)} - x^\star\|$ is no greater than some positive threshold at some k, the algorithm terminates and we accept $x^{(k)}$ as a solution. However, since the true $x^\star$ is unknown, we need to estimate $\|x^{(k)} - x^\star\|$.

9.2 Analysis on Gradient Algorithm

- Given an initial guess $x^{(0)}$, define the set $\mathcal{S} := \{x \in \mathbb{R}^n | f(x) \leq f(x^{(0)})\}$. It is said that ∇f is Lipschitz continuous on $\mathcal{S}$ with constant $L > 0$ if $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$ for all $x, y \in \mathcal{S}$. If f is twice continuous differentiable and S is compact, then L exists.
- Suppose that L exists. For the gradient algorithm, consider an arbitrary $\epsilon > 0$. If the step size $s^{(0)}, s^{(1)}, \ldots$ are chosen in the interval $(\epsilon, \frac{2}{L})$, then $\|\nabla f(x^{(k)})\| \to 0$ as $k \to \infty$.
  - This means that the algorithm converges to a stationary point (which can be a local minimum, a local maximum, or a saddle point).
  - If $f(x)$ is convex, then $\nabla f(x^\star)$ is zero if and only if $x^\star$ is the global minimum. Hence, gradient algorithm always converges to a global minimum of a convex function if $s^{(k)}$ is small for all k.

10 Applications

10.1 Low-Rank Matrix Approximation

Given a matrix $A \in \mathbb{R}^{m\times n}$, consider the problem of finding a low-rank matrix $B \in \mathbb{R}^{m\times n}$ that best approximates A.

- This problem can be formulated as $\min_{B\in\mathbb{R}^{m\times n}} \|A - B\|_{2 \text{ or } F}$ subject to $\text{Rank}(B) \leq k$.
- Eckart-Young-Mirsky theorem:
  - For a given $k \leq \min(m, n)$, define $A_k := \sum_{i=1}^k \sigma_i u^{(i)} v^{(i)\top}$ constructed with the top k singular values of A and the corresponding left/right singular vectors. A_k has rank at most k. Intuitively, we "chop off" the smaller singular values starting from the $k+1$-th largest.
  - $B = A_k$ is an optimal solution to both optimization problems (Frobenius or ℓ_2-induced norm).
  - Suppose that $k < \text{Rank}(A)$. The optimal solution is unique if and only if $\sigma_k \neq \sigma_{k+1}$, i.e., the k-th largest singular value of A is not equal to the $k+1$.
- The relative Frobenius norm approximation error $e_k := \frac{\|A-A_k\|_F^2}{\|A\|_F^2}$ is equal to $\frac{\sigma_{k+1}^2+\ldots+\sigma_r^2}{\sigma_1^2+\ldots+\sigma_r^2}$, where $r = \text{Rank}(A)$.
- The relative ℓ_2-induced norm approximation error $\frac{\|A-A_k\|_2}{\|A\|_2}$ is equal to $\frac{\sigma_{k+1}}{\sigma_1}$.

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10.2 Principal Component Analysis (PCA)

- Given points $x^1, \ldots, x^m$, first center data points to $\tilde{x}^1, \ldots, \tilde{x}^m$ by subtracting $\sum_{i=1}^m x^i$.
- Compute the left singular vectors $v^1, \ldots, v^m$.
- Most variation is along v^1 (explains $\sigma_1^2 / \sum_i \sigma_i^2$), second most along v^2, etc.

10.3 Robust PCA

We aim to decompose $Y \in \mathbb{R}^{m \times n}$ as the sum of a low-rank matrix $X \in \mathbb{R}^{m \times n}$ and a sparse (most entries are zero) matrix $Z \in \mathbb{R}^{m \times n}$. To achieve this, we can solve the optimization problem

$$\min_{X, Z \in \mathbb{R}^{m \times n}} \text{Rank}(X) + \lambda \, \text{Card}(Z) \quad \text{subject to} \quad Y = X + Z,$$

where $\text{Card}(Z)$ is the number of non-zero entries in Z and $\lambda > 0$ is a regularization coefficient.

The above problem is non-convex. To this end, we can solve the following convex problem as a surrogate:

$$\min_{X, Z \in \mathbb{R}^{m \times n}} \|X\|_* + \lambda \sum_{i=1}^m \sum_{j=1}^n |Z_{ij}| \quad \text{subject to} \quad Y = X + Z.$$

10.3.1 Video Processing

Consider the problem of identifying moving objects in a video.

- Suppose that the video has q frames, $V^{(1)}, \ldots, V^{(q)}$, where the (i, j) entry $V^{(k)}$ is the color value of the (i, j) pixel at frame k for $k = 1, \ldots, q$.
- We compose each frame $V^{(k)}$ into the foreground $V_f^{(k)}$ (moving objects) and the background $V_b^{(k)}$ (changes slowly). Assuming that the moving object occupies a small portion of the video, then $V_f^{(k)}$ is sparse.
- Now we vectorize each $V^{(k)}$ and concatenate them into a single matrix as $Y = \begin{bmatrix} \text{vec}(V^{(1)}) & \ldots & \text{vec}(V^{(k)}) \end{bmatrix}$. We can then decompose the known matrix Y as $Y = X + Z$, where X is low-rank and Z is sparse.
  - X takes on the role of $\begin{bmatrix} \text{vec}(V_b^{(1)}) & \ldots & \text{vec}(V_b^{(k)}) \end{bmatrix}$.
  - Z takes on the role of $\begin{bmatrix} \text{vec}(V_f^{(1)}) & \ldots & \text{vec}(V_f^{(k)}) \end{bmatrix}$.
- We can then solve this decomposition via robust PCA.

10.4 Matrix Completion

- Consider a matrix $X^\star = \mathbb{R}^{m \times n}$ whose entries are unknown but is known to be low rank. Assume that we measure the entries $X_{ij}^\star$ only when (i, j) belongs to some given set $\mathcal{S}$.
- To estimate $X^\star$ using the measurements, we can find the lowest-rank X whose (i, j) entries match the measurements by solving for the optimization problem

$$\min_{X \in \mathbb{R}^{m \times n}} \text{Rank}(X) \quad \text{subject to} \quad X_{ij} = X_{ij}^\star, \ \forall (i, j) \in \mathcal{S}.$$

- This problem is non-convex due to the discrete rank function in the objective.
  Over the restricted space $\{X \in \mathbb{R}^{m \times n} \mid \|X\|_2 \leq 1\}$, a convex relaxation is

$$\min_{X \in \mathbb{R}^{m \times n}} \|X\|_* \quad \text{subject to} \quad X_{ij} = X_{ij}^\star, \ \forall (i, j) \in \mathcal{S}.$$

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10.5 Compressed Sensing

- Let $x^\star \in \mathbb{R}^n$ denote some states of some system. We want to know $x^\star$ but can only measure $b := Ax^\star \in \mathbb{R}^m$ for some $m \times n$ matrix A. When $m < n$, the linear system is underdetermined.
- Suppose that $x^\star$ is known to be sparse. Then $x^\star$ can be estimated via the optimization problem

$$\min_x \|x\|_0 \quad \text{subject to} \quad Ax = b.$$

- This problem is non-convex, but can be approximated with its convex relaxation over the restricted space of $-1 \leq x \leq 1$:

$$\min_x \|x\|_1 \quad \text{subject to} \quad Ax = b,$$

  which can be reformulated as an LP

$$\min_{x,t \in \mathbb{R}^n} \mathbf{1}_n^\top t \quad \text{subject to} \quad Ax = b,\ -t \leq x \leq t,$$

  where $\mathbf{1}_n$ denotes the n-dimensional all-one column vector.
- Suppose that our measurements are noisy, i.e., $\hat{b} = Ax + w$ where w is random (often Gaussian). Then, the problem we should solve is

$$\min_x \|w\|_2^2 + \lambda \|x\|_1 \quad \text{subject to} \quad Ax + w = b,$$

  where $\lambda > 0$ is a user-defined balancing constant. This is a constrained LASSO problem that can be reformulated as a QP (see Section 6.6).

10.6 Power Network Identification with Adversarial Sensor Injections

- Power networks can be represented as a graph, where each node n_i is a generator or a load and each edge e_{ij} connects node n_i to node n_j.
- Each node n_i has phase θ_i and each edge e_{ij} has a reactance $z_{ij} \in \mathbb{R}$.
- The power flow on each edge is $p_{ij} = \frac{\theta_i - \theta_j}{z_{ij}}$.
- Due to the graph structure, $p_{ij} = -p_{ji}$. Furthermore, the power at each node n_i is $p_i = \sum_{j \in \mathbb{N}_i} p_{ij}$, where $\mathbb{N}_i$ is the set of indices of all nodes n_j that are connected to n_i via an edge e_{ij}.
- We have node measurements $\hat{p}_i$ for all $i \in \hat{\mathbb{N}}$ where $\hat{\mathbb{N}}$ is the set of measured node indices. Here, Here, $\hat{p}_i = p_i + w_i + v_i$, where w_i is sensor noise (usually random, dense, but small) and v_i is potential adversarial injection (usually large but sparse).
- We also have edge measurements $\hat{p}_{ij}$ for all $(i,j) \in \hat{\mathbb{E}}$ where $\hat{\mathbb{E}}$ is the set of measured edge indices. Here, $\hat{p}_{ij} = p_{ij} + w_{ij} + v_{ij}$, where w_{ij} is sensor noise and v_{ij} is potential adversarial injection.
- We assume that only a small number of nodes and edges are subject to adversarial injection.
- Given the (noisy and potentially adversarially perturbed) sensor measurements, we want to estimate the power network status (i.e., the true p_i and p_{ij}). To achieve so, we can solve the LASSO problem

$$\begin{aligned} \min_{w,v,\theta} \quad \|w\|_2^2 + \lambda \|v\|_1 \quad \text{subject to} \quad & \hat{p_i} = \sum_{j \in N_i} \frac{\theta_i - \theta_j}{z_{ij}} + w_i + v_i, \\ & \hat{p_{ij}} = \frac{\theta_i - \theta_j}{z_{ij}} + w_{ij} + v_{ij}, \\ & \theta_1 = 0. \end{aligned}$$

  Here, each $\hat{p}_i$ and each $\hat{p}_{ij}$ is measured, each z_{ij} is known, w and v are vectors that contains all w_i, w_{ij} and v_i, v_{ij} respectively, θ is a vector that contains all node phases, and $\lambda > 0$ is a hyperparameter that adjusts the sparsity of the solution.

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